

Practical 8

* Aim:- Implement sorting algorithms such as quick sort, merge sort, and heap sort, and compare their performance.

* Theory:- Sorting is a technique which we perform on data structure in-order to arrange data elements in ascending and descending order. There are different methods that can be used to arrange data:-

- (i) Quick Sort
- (ii) Merge Sort
- (iii) Heap Sort

(i) Quick Sort:- Quick sort is one of the fastest sorting algorithms used. It works on the divide and conquer technique. In this method, one element is chosen as a pivot. The array is divided into two parts: one containing elements smaller than the pivot and other containing elements larger than the pivot. These parts are then recursively sorted.

Quick sort generally performs very well for large datasets and requires less memory compared to other sorting methods. However, if the pivot is not chosen properly, its performance may become slow in the worst case.

(ii) Merge Sort:- Merge sort also follows the divide and conquer approach. It divides the array

into two equal parts until each subarray contains only one element. Then these subarrays are merged back together in a sorted manner.

Merge sort always gives consistent performance because its time complexity remains the same in all cases. It is a stable sorting algorithm, meaning it preserves the relative order of equal elements. The main drawback of Merge Sort is that it requires extra memory to store temporary arrays during the merging process.

③ Heap Sort :- Heap Sort is based on a binary heap data structure. In this, the elements are first arranged in the form of the max heap, where the largest element is at the root. The largest element is repeatedly removed and placed at the end of the array, and the heap is restructured at each time.

Heap sort does not require additional memory and provides guaranteed performance in all cases. However, it is generally slower than Quick Sort due to more comparisons and complex operations.

* Comparison Table :-

Parameter	Quick Sort	Merge Sort	Heap Sort
① Sorting Technique	Divide & Conquer	Divide & Conquer	Heap (Binary Heap)

	Quick Sort	Merge Sort	Heap Sort
② Best Case Time	$O(n \log n)$	$O(n \log n)$	$O(n \log n)$
③ Average Case Time	$O(n \log n)$	$O(n \log n)$	$O(n \log n)$
④ Worst Case Time	$O(n^2)$	$O(n \log n)$	$O(n \log n)$
⑤ Space Complexity	$O(\log n)$	$O(n)$	$O(1)$
⑥ Stability	Not Stable	Stable	Not stable
⑦ Extra Memory Needed	Very low	High	None
⑧ Speed	Very fast	Moderate	Slower than Quick Sort
⑨ Suitable for	Large data set	Linked list, external sorting	Memory-Limited system

* Conclusion:- Quick Sort, Merge Sort, and Heap Sort were implemented and analyzed to compare their performance. All three algorithms are efficient and have an average time complexity of $O(n \log n)$, making them suitable for large data sets.

Among these algorithms, Quick sort is the most efficient to use because it requires less memory and performs faster for most real-world inputs. Merge Sort provides stable and consistent performance but uses extra memory, while heap Sort guarantees $O(n \log n)$ time in all cases but is generally slower due to more comparisons.